October 2025

Christina M. Qiu

Address: Department of Economics
Yale University
New Haven, CT 06520-8268

Telephone: +1 (973) 906-5296

E-mail: christina.qiu@yale.edu

Website: https://christinamqiu.github.io/

Citizenship: United States

Fields of Concentration:
Primary Field(s): International Economics, Development Economics
Secondary Field(s): Macroeconomics

Qualifying Examinations Completed:
2022 (Oral): International Trade (*with distinction*), Macroeconomics

Dissertation Title: *Essays on Technology and Spatial Economics*

Committee:
Professor Costas Arkolakis (Chair)
Professor Mayara Felix
Professor Joel Flynn
Professor Mushfiq Mobarak

Education:
Ph.D., Economics, Yale University, 2026 (expected)
M.Phil., Economics, Yale University, 2022
M.A., Economics, Yale University, 2021
M.Sc., Global Governance and Diplomacy (*with distinction*), Oxford University, 2020
B.A., Applied Mathematics (*awarded magna cum laude*), Harvard College, 2019

Fellowships, Honors and Awards:
Graduate Research Fellowship, National Science Foundation, 2020-2025
Carl Arvid Anderson Prize Fellowship ($19,000), Yale University, 2024-2025
Ryoichi Sasakawa Young Leaders Fellowship ($4,000), Yale University, 2022-2024
Outstanding Academic Achievement Prize (*highest scorer in exams & dissertation*), Oxford University, 2020
Clarendon Fellowship (*full scholarship*), Oxford University, 2019-2020
Phi Beta Kappa (*top 10% of class*), Harvard College, 2019
John Dunlop Thesis Prize (*honorable mention*), Harvard Kennedy School, 2019

Kenneth I. Juster Fellowship, Harvard University, 2018

Distinction in Teaching Award (*avg. score 4.5/5 or higher*), Harvard College, 2018

Region 1 Mark of Excellence Award Winner (*in General Column Writing*), Society of Professional Journalists, 2017

Cyrilly Abels Short Story Prize (*2nd place*), Department of English, Harvard College, 2016

U.S. Presidential Scholar in the Arts (*writing*), U.S. Department of Education & National YoungArts Foundation, 2015

Research Grants:

Alfred P. Sloan Foundation Minor Research Grants in Mesoeconomics ($5000), "Eliciting the Beliefs of Supply Chain Managers", 2024-2025

Teaching Experience:

Fall 2024, Teaching Assistant to Ana Cecilia Fieler, Growth & Macroeconomics (G), Yale University

Spring 2024, Grader to Kaivan Munshi, Social Networks & Economic Development (G), Yale University

Summers 2022 & 2023, Supervisor at Scarf Research Program for Yale Undergraduates

Summer 2021, Teaching Assistant to Jonathan Hawkins, Intermediate Microeconomics (U), Yale College Summer Session

Spring 2018, Classroom Assistant to Jeremy Hahn, Linear Algebra & Differential Equations (U), Harvard College

Research Experience:

Research Assistant to Teresa Fort, Dartmouth University, Fall 2023 – Spring 2024

Research Assistant to Costas Arkolakis, Yale University, Summer 2022

Research Assistant to Orazio Attanasio, Yale University, Summer 2022

Data Science Fellow for Bureau of Labor Statistics, Civic Digital Fellowship, Summer 2019

Publications:

Qiu, C. M. (2022). Regionalized liquidity: A cross-country analysis of mobile money deployment and inflation in developing economies. *World Development*, 152, 105781.

Working Papers:

"Endogenous Transfer Networks Under Spatial Risk", (November 2025), *Job Market Paper*

"AI, Scale, and Skills: A Quantitative Task-Based Theory of Automation" with Danial Lashkari, Wensu Li, and Neil Thompson, (July 2025)

Work In Progress:

"Dynamic Management of Supplier Capital" with Will Jianyu Lu, (2025)

"Geographic Diversification in Home Insurance Markets" with Xianglong Kong and Mark Walker, (2025)

Seminar and Conference Presentations:

2024: G-TAP Annual Conference
2023: European Economic Association, Urban Economics Association

Referee Service:
World Development, Journal of International Economics

Languages:
English (native), French (intermediate)

References:

Prof. Costas Arkolakis
Yale University
Department of Economics
New Haven, CT 06520
Phone: (203) 432-3527
costas.arkolakis@yale.edu

Prof. Mayara Felix
Yale University
Department of Economics
New Haven, CT 06520
Phone: (413) 437-0105
mayara.felix@yale.edu

Prof. Joel Flynn
Yale University
Department of Economics
New Haven, CT 06520
Phone: (203) 432-0807
joel.flynn@yale.edu

Prof. Mushfiq Mobarak
Yale University
School of Management
New Haven, CT 06520
Phone: (203) 432-5787
ahmed.mobarak@yale.edu

Dissertation Abstract

My dissertation examines the economic impacts of disruptive new technologies on risk-sharing arrangements, wages, and welfare.

Endogenous Transfer Networks Under Spatial Risk [Job Market Paper]

Spatial risk—or the presence of location-specific shock—is a major source of uncertainty in developing economies. With limited access to formal financial markets, households rely on informal transfers to manage income volatility. New digital payment systems reduce the cost of money transport over long distances, which reshapes the value of risk-sharing ties. Pioneered by Safaricom's M-Pesa in 2008 across sub-Saharan Africa, mobile money permits cashless transfers for households without connections to a bank account. How does mobile money reshape transfer networks and alter their capacity to manage spatial risk across heterogeneous households? What welfare gains arise when endogenous network changes are taken into account?

To answer these questions, I develop a model in which households, differentiated by productivity and location, form transfer networks over space. Pairs choose symmetric partnership levels that specify income shares transferred if requested. Each household experiences idiosyncratic and spatially correlated income shocks. After the shocks, households may request a payment from a partner subject to distance-dependent transport costs. The model yields aggregate transfer flows in a pairwise-stable equilibrium. In this setting, households smooth consumption in part by partnering with others who are themselves well-diversified—a new strategic consideration in spatial models.

The model predicts that by reducing money transport costs, mobile money facilitates connections with high-productivity partners. Yet networks adjust endogenously, leaving different types of households unevenly exposed to risk. Persistent shocks that trigger network changes sever profitable partnerships for low-productivity households but strengthen the networks of high-productivity ones. Both effects are amplified by mobile money. Using Tanzanian household panel data and a shift-share instrument exploiting variation in village "tech-savviness", I show that mobile money increases partners' education levels by 2.8 years on average. It also reduces transfer distance by 113.8 km for low-education households hit by unexpected negative shocks involving capital loss, while high-education households increase transfer volumes five-fold.

I calibrate the model to match post-2008 aggregate transfer flows in Tanzania, a period of substantial increase in the spatial diversification of money transfer sources. On average, mobile money increased household welfare by 2.2 percentage points, and gains were progressive. Quantitatively, endogenous network readjustments in transfer partnerships account for 32.2% of the gains. Despite the fragility of their transfer partnerships, low-productivity households enter risk-sharing arrangements as net receivers, while high-productivity households participate primarily for insurance and act as net givers.

AI, Scale, and Skills: A Quantitative Task-Based Theory of Automation, with Danial Lashkari, Wensu Li, and Neil Thompson.

This paper develops a quantitative task-based theory of automation that allows a direct mapping to empirical measures of AI exposure and AI automation costs. The theory accounts for the increasing returns to scale property of AI, where fixed training costs often outweigh variable inference costs. Firms' automation decisions across tasks are shaped not only by the relative productivity of machines to labor, but also by patterns of *scale advantage*, i.e., a task's output to fixed cost ratio. We derive task, firm, and economy-wide elasticities of substitution between labor demand and capital prices under firms' endogenous adoption of AI with fixed costs. We directly estimate the training and inference costs of AI using robust empirical relationships between AI performance and its computational inputs. To do so, we employ a worker survey to uncover information on relevant task characteristics, including *complexity*, i.e., a task's number of potential outcomes, and *required accuracy*, i.e. the minimum proportion of "successful" attempts a worker can make to be considered qualified to conduct a task. We calibrate our baseline model to AI adoption rates published by the Census Bureau in 2023. Quantitatively, labor becomes *less* substitutable with AI when capital prices decline, as remaining workers are highly productive and become increasingly difficult to replace.

October 2025

Dynamic Management of Supplier Capital, with Will Jianyu Lu

This paper develops a theory of *supplier capital*, treating the geographic distribution of a firm's suppliers as a stock subject to depreciation. The framework rests on two principles: first, supplier relationships show persistence over time, and second, a supplier set's geographic properties affect input price levels and volatility. In the model, each location offers a continuous menu of supplier qualities indexed by volatility. Firms adjust supplier capital along two margins: deepening relationships within a location by accessing lower-volatility suppliers, and diversifying across locations by creating new supplier ties. Diversification substitutes for deepening, while deepening displays increasing returns. The model generates state-dependent shifts in firm focus between profitability and resilience, as well as path-dependent firm costs of deepening supplier capital. Moreover, endogenous profitability shapes risk attitudes, leading firms to organize their supply chains to be more resilient as they mature.